Mushir Uz Zaman

Mathematics with Computer Science Graduate

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M.Sc. Mathematics with Computer Science graduate from Jamia Millia Islamia, New Delhi, with a completed B.Sc. in Mathematics. Academic preparation combines rigorous mathematical reasoning with C and Java programming, data structures, operating systems, database management systems, and SQL. Interested in AI training and review roles that require careful multi-step analysis, code interpretation, scientific reasoning, and clear technical feedback in Hindi and professional English.

CORE STRENGTHS AND COURSEWORK

Strongest areas, in order: 1. Mathematics 2. Coding & Computer Science.

Mathematics: Real Analysis, Abstract Algebra, Topology, Linear Algebra, Differential Equations and Applications, Functional Analysis, Complex Analysis, Differential Geometry, Differentiable Manifolds, Wavelet Analysis, and Numerical Analysis. Coursework includes measure and integration, convergence, groups and rings, vector spaces, eigenvalues, boundary-value problems, Fourier methods, and numerical approximation.

Numerical & Mathematical Reasoning: Root-finding, interpolation, numerical integration, Runge-Kutta methods, finite differences, stability analysis, approximation, and error estimation. Relevant foundation for checking multi-step solutions, assumptions, and numerical accuracy.

Coding: C programming fundamentals including control flow, arrays, strings, pointers, functions, recursion, structures, and file handling. Java coursework includes object-oriented programming, classes, inheritance, polymorphism, interfaces, collections, and exception handling.

Computer Science: Data structures and algorithms in C/Java: arrays, linked lists, stacks, queues, trees, graphs, searching, sorting, BFS/DFS, and time/space complexity. Operating Systems: processes, scheduling, synchronization, deadlocks, memory management, and file systems. DBMS: relational models, ER diagrams, normalization, transactions, and SQL.

Physics: Undergraduate study in mechanics, motion, waves, optics, electricity and magnetism, measurement, and mathematical modelling of physical systems. Supports quantitative and scientific reasoning.

Chemistry: Undergraduate study in physical, organic, and inorganic chemistry, with scientific observation, quantitative reasoning, and laboratory-based concepts.

Communication: Hindi (native) and professional English for reading, writing, explanation, and structured technical communication.

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PROJECTS

Study of Three Generalized Fixed Point Theorems

Academic Mathematics Project

- Studied three generalized fixed point theorems with focus on mappings, metric spaces, convergence conditions, and proof-based reasoning.
- Developed structured understanding of functional analysis concepts and the conditions leading an iterative sequence toward a stable solution.

Personal Academic Portfolio

Portfolio: mushir-portfolio.pages.dev

- Built and published a responsive personal portfolio covering academic background, coursework, project work, online certificates, coding practice, and professional profiles.
- Designed for phone, tablet, and laptop use with interactive academic-topic and practice sections.

EDUCATION

M.Sc. Mathematics with Computer Science

Jamia Millia Islamia, New Delhi | Completed

Academic focus: advanced mathematics, numerical analysis, C/Java programming, data structures, operating systems, web designing, and database management systems.

B.Sc. in Mathematics

Vinoba Bhave University, Jharkhand | Completed

Mathematics as the main subject, with Physics and Chemistry electives; foundation in quantitative problem-solving and scientific study.

ONLINE COURSES AND CERTIFICATES

Python | Kaggle Learn

Course certificate completed. Topics include functions, conditionals, loops, lists, strings, dictionaries, and libraries.

Intro to Machine Learning | Kaggle Learn

Course certificate completed. Topics include model building, validation, underfitting/overfitting, and random forests.

Python and SQL Practice | HackerRank

Programming and SQL problem-solving practice through public HackerRank profile.

Algorithmic Problem Solving | LeetCode

Practice solving coding questions and algorithmic problems.

TECHNICAL SKILLS

Mathematics · C · Java · Python · SQL · Data Structures · Algorithms · DBMS · Operating Systems · HTML · CSS · JavaScript · Machine Learning

LANGUAGES

Hindi — Native | English — Professional proficiency

ONLINE PROFILES

GitHub

LinkedIn

Kaggle

HackerRank

LeetCode